

# Chapter 5: Building Your Research

## From Concepts to Testable Studies

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### 1 Overview

#### Before You Begin

This chapter bridges the gap between understanding experimental design concepts and actually planning your own research. You'll learn to turn abstract ideas into concrete, testable studies — the foundation of your Research Plan assessment.

This chapter covers:

- Operationalisation: Making variables measurable
- Writing hypotheses: One-tailed vs two-tailed choices
- Design decisions: Between-subjects vs within-subjects trade-offs
- Connecting design to analysis: Why your design determines your test
- Method section structure: What to include and why

By the end of this reading, you should feel prepared to start planning your Research Plan.

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### 2 Part 1: What is Operationalisation?

#### 2.1 The Gap Between Ideas and Measurement

Research starts with ideas: *Does stress affect memory? Does eye contact build trust? Does hydration improve thinking?*

These are interesting questions, but they're not yet testable. To test them, we need to answer:

- What exactly do we mean by “stress” or “memory” or “trust”?
- How will we create different levels of our independent variable?
- How will we measure the dependent variable?

This is where operationalisation comes in.

**Operationalisation:** The process of turning an abstract concept into a concrete, measurable variable with a specific procedure for manipulation or measurement.

## 2.2 Why Operationalisation Matters

Good operationalisation enables:

1. **Replication:** Other researchers can repeat your study exactly
2. **Communication:** Readers understand precisely what you did
3. **Validity:** Your measure actually captures what you claim to measure
4. **Control:** You know exactly what differs between conditions

Poor operationalisation leads to:

1. **Ambiguity:** “We measured mood” — how? With what?
2. **Confounds:** Multiple things differ between conditions
3. **Irreplicability:** No one can reproduce your findings
4. **Invalid conclusions:** You measured something, but not what you think

## 2.3 Operationalising the Independent Variable

The IV is what you manipulate. You need to specify exactly how the conditions differ.

**Vague:** “Participants were either stressed or not stressed”

**Operationalised:** “Participants in the stress condition completed a timed mental arithmetic task (counting backwards from 2,043 by 17s) while being observed by an experimenter who maintained a neutral expression. Participants in the control condition sat quietly for the same duration (3 minutes) without any task.”

Notice how the operationalised version tells you:

- What the stressor was (timed arithmetic)
- Specific parameters (starting number, subtraction amount)
- The social element (observed, neutral expression)
- The control condition (matched for time, no task)
- Duration (3 minutes)

## 2.4 Operationalising the Dependent Variable

The DV is what you measure. You need to specify the exact measurement procedure.

**Vague:** “We measured their memory”

**Operationalised:** “Memory was assessed using a free recall task. Participants were shown 20 words (10 concrete nouns, 10 abstract nouns) for 3 seconds each. After a 30-second filled delay (counting backwards from 50), participants wrote down as many words as they could remember in 2 minutes. The dependent variable was the total number of correctly recalled words (maximum = 20).”

This version specifies:

- The type of memory test (free recall)
- The stimuli (20 words, types specified)
- Presentation timing (3 seconds each)

- The retention interval (30 seconds, filled)
- The recall procedure (written, 2 minutes)
- The scoring method (total correct, max = 20)

## 3 Part 2: The Three Research Plan Topics

Your Research Plan assessment requires you to design an experiment on one of three topics. Let's work through the operationalisation challenges for each.

### 3.1 Topic 1: Font Difficulty → Memory

**The Research Question:** Does reading text in a harder-to-read font improve memory retention?

**Background:** Diemand-Yauman et al. (2011) proposed that “desirable difficulties” — challenges that require more cognitive effort during learning — can enhance long-term retention. Reading disfluent fonts requires more effort, which might deepen processing.

#### 3.1.a Operationalising “Font Difficulty”

You need two conditions that differ ONLY in font readability:

| Condition        | Font Specification                         |
|------------------|--------------------------------------------|
| Fluent (Easy)    | Arial, 16-point, black on white            |
| Disfluent (Hard) | Comic Sans MS, 12-point, 75% grey on white |

The key principle: Keep everything else constant (same words, same screen, same instructions) so that font is the only thing that differs.

#### Alternative operationalisations:

- Fluent vs Italic versions of the same font
- Standard vs reduced contrast
- Clear vs slightly blurred text

#### Potential Confound

If your fonts differ in more than just readability (e.g., one is much smaller, or associated with different contexts), you can't be sure which difference caused any effect.

#### 3.1.b Operationalising “Memory”

Common options:

| Measure     | Description                             | Pros                           | Cons                             |
|-------------|-----------------------------------------|--------------------------------|----------------------------------|
| Free recall | Write down everything remembered        | Sensitive to deeper processing | Floor/ceiling effects possible   |
| Recognition | Multiple choice: "Was this word shown?" | Easy to score                  | May not detect subtle effects    |
| Cued recall | Given category, recall specific items   | Balanced difficulty            | Requires careful stimulus design |

Your choice should be justified based on prior research or theoretical considerations.

### 3.2 Topic 2: Hydration → Cognitive Performance

**The Research Question:** Does drinking water improve cognitive task performance?

**Background:** Edmonds & Burford (2009) found that children who drank water before cognitive tests performed better. Dehydration may impair attention and processing speed.

#### 3.2.a Operationalising "Hydration"

Options for your IV manipulation:

| Approach            | Water Condition                     | Control Condition           | Considerations                     |
|---------------------|-------------------------------------|-----------------------------|------------------------------------|
| Simple manipulation | Drink 300ml water                   | No drink                    | Confound: act of drinking vs water |
| Placebo control     | Drink 300ml water                   | Sip water (no swallowing)   | Controls for drinking action       |
| Standardised        | Drink 250ml after 2-hour abstinence | 2-hour abstinence, no drink | Controls prior hydration state     |

#### i Design Consideration

Consider whether participants knowing they're in a "hydration" study might create expectation effects. If people believe water helps cognition, their performance might improve regardless of actual hydration.

#### 3.2.b Operationalising "Cognitive Performance"

Options include:

- **Attention tasks:** Simple/choice reaction time, sustained attention (continuous performance task)

- **Working memory:** Digit span, n-back task
- **Processing speed:** Symbol-digit substitution, pattern comparison
- **Executive function:** Stroop task, Wisconsin Card Sorting Test

Your choice should match the theoretical mechanism. If hydration affects attention, use an attention task.

### 3.3 Topic 3: Eye Contact → Perceived Trustworthiness

**The Research Question:** Does eye contact duration affect how trustworthy a person is perceived to be?

**Background:** Argyle & Dean (1965) established that eye contact is a fundamental component of social interaction. Subsequent research has linked direct gaze to perceptions of competence, confidence, and trustworthiness.

#### 3.3.a Operationalising “Eye Contact”

Options for manipulation:

| Approach    | High Eye Contact         | Low Eye Contact            | Considerations                    |
|-------------|--------------------------|----------------------------|-----------------------------------|
| Photographs | Eyes looking at camera   | Eyes looking slightly away | Maximum control, less realistic   |
| Videos      | 80% direct gaze          | 20% direct gaze            | More realistic, harder to control |
| Duration    | 5 seconds of direct gaze | 1 second of direct gaze    | Tests duration specifically       |

Using photos or videos allows tight control — you can be sure the only difference is eye contact. Live interactions are more ecologically valid but introduce many confounds (voice, body language, natural variation).

#### 3.3.b Operationalising “Trustworthiness”

Options:

| Measure             | Example                                 | Pros           | Cons                         |
|---------------------|-----------------------------------------|----------------|------------------------------|
| Single rating scale | “How trustworthy is this person?” (1-7) | Simple, direct | May not capture full concept |
| Multiple-item scale | Trust subscale from established measure | More reliable  | Longer, may be redundant     |
| Behavioural measure | “Would you lend this person £10?”       | Face valid     | Binary data less sensitive   |

| Measure       | Example                              | Pros                    | Cons                  |
|---------------|--------------------------------------|-------------------------|-----------------------|
| Economic game | How much would you invest with them? | Behavioural, continuous | Complex to administer |

Multi-item scales are generally more reliable than single items. Consider using an established trustworthiness measure if one exists.

## 4 Part 3: Writing Hypotheses

### 4.1 Review: Null and Alternative Hypotheses

Every statistical test compares your data against a null hypothesis.

**Null Hypothesis ( $H_0$ ):** There is no effect — any observed difference is due to chance.

**Alternative Hypothesis ( $H_1$ ):** There IS an effect — the observed difference is not due to chance.

For the font study:

- $H_0$ : Font type has no effect on memory ( $\mu_{\text{fluent}} = \mu_{\text{disfluent}}$ )
- $H_1$ : Font type does affect memory ( $\mu_{\text{fluent}} \neq \mu_{\text{disfluent}}$ )

### 4.2 One-Tailed vs Two-Tailed Hypotheses

The alternative hypothesis can be:

**Two-tailed (non-directional):** Predicts there IS a difference, without specifying which direction.

$H_1$ : Memory scores will differ between fluent and disfluent font conditions.

**One-tailed (directional):** Predicts the specific direction of the difference.

$H_1$ : Memory scores will be higher in the disfluent font condition than the fluent font condition.

### 4.3 When to Use Each

| Use Two-Tailed When...                | Use One-Tailed When...                          |
|---------------------------------------|-------------------------------------------------|
| Effect could reasonably go either way | Strong theoretical basis for direction          |
| Prior literature is mixed             | Consistent prior findings support one direction |
| You're exploring a new question       | Replicating established effect                  |

| Use Two-Tailed When...                | Use One-Tailed When...                             |
|---------------------------------------|----------------------------------------------------|
| You want to detect unexpected effects | You're focused on confirming a specific prediction |

## 4.4 Justifying Your Choice

For your Research Plan, you must justify why you chose one-tailed or two-tailed:

### Example two-tailed justification:

"A two-tailed hypothesis was used because prior research on disfluency effects has produced mixed results. While Diemand-Yauman et al. (2011) found memory benefits, subsequent replications have been inconsistent (Rummer et al., 2016). Given this uncertainty, we test for any difference rather than predicting a specific direction."

### Example one-tailed justification:

"A one-tailed hypothesis was used because the desirable difficulties framework (Bjork & Bjork, 2011) and the original disfluency research (Diemand-Yauman et al., 2011) both predict that harder-to-read fonts should enhance memory. We therefore predict that recall will be higher in the disfluent condition."

### ! Assessment Requirement

Your Research Plan **MUST** include justification for your hypothesis direction choice. This demonstrates critical thinking about the research literature and your theoretical position.

## 4.5 Writing Complete Hypotheses

A complete hypothesis statement includes:

1. The IV and its levels
2. The DV and how it's measured
3. The predicted relationship

### Complete two-tailed:

"Participants who read a word list in a disfluent font (Comic Sans MS, 12pt, 75% grey) will show different free recall scores compared to participants who read the same word list in a fluent font (Arial, 16pt, black)."

### Complete one-tailed:

“Participants who read a word list in a disfluent font (Comic Sans MS, 12pt, 75% grey) will recall significantly more words than participants who read the same word list in a fluent font (Arial, 16pt, black).”

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## 5 Part 4: Design Decisions — Between vs Within

### 5.1 The Fundamental Trade-off

You’ve chosen your topic, operationalised your variables, and written your hypotheses. Now you must decide how to assign participants to conditions.

**Between-Subjects Design:** Different participants in each condition. Each person experiences only ONE level of the IV.

**Within-Subjects Design:** Same participants in all conditions. Each person experiences EVERY level of the IV.

### 5.2 Between-Subjects Design

**How it works:**

- Randomly assign 30 participants to Condition A
- Randomly assign 30 different participants to Condition B
- Compare the mean scores of each group

**Advantages:**

1. **No order effects:** Participants can’t be influenced by having done another condition first
2. **No practice effects:** Performance doesn’t improve/decline across conditions
3. **No demand characteristics from comparison:** Participants don’t try to be “consistent” or “different”
4. **Simpler procedure:** Each participant does one thing

**Disadvantages:**

1. **Individual differences add noise:** Smart people in one group inflate that mean
2. **Needs more participants:** Twice as many people for two conditions
3. **Pre-existing group differences:** Even with randomisation, groups might differ

### 5.3 Within-Subjects Design

**How it works:**

- All 30 participants complete Condition A
- Same 30 participants complete Condition B
- Compare each person’s scores across conditions

**Advantages:**

1. **Controls individual differences:** Each person is their own control



2. **More statistical power:** Less “noise” makes real effects easier to detect
3. **Fewer participants needed:** Same people generate data for all conditions
4. **Can detect small effects:** Removing individual differences un.masks subtle differences

**Disadvantages:**

1. **Order effects:** First condition might affect second
2. **Practice effects:** Performance changes with repeated testing
3. **Fatigue effects:** Performance declines with longer sessions
4. **Demand characteristics:** Participants might guess what you expect

## 5.4 Design Decisions for Each Topic

### 5.4.a Font Difficulty → Memory

| Design  | Implementation                                | Key Concern                              |
|---------|-----------------------------------------------|------------------------------------------|
| Between | Group A reads fluent, Group B reads disfluent | Individual differences in memory ability |
| Within  | Everyone reads BOTH font versions             | Memory for first list affects second     |

**Typical choice:** Between-subjects

**Reasoning:** If participants read the same words in both fonts, memory from the first reading will contaminate the second. If we use different word lists, we can’t be sure the lists are equivalent in difficulty.

### 5.4.b Hydration → Cognitive Performance

| Design  | Implementation                               | Key Concern                                       |
|---------|----------------------------------------------|---------------------------------------------------|
| Between | Group A drinks water, Group B doesn’t        | Individual differences in cognitive ability       |
| Within  | Same person tested hydrated and not hydrated | Requires two sessions or dehydration manipulation |

**Typical choice:** Between-subjects (simpler) or within-subjects with two sessions (more powerful but logistically harder)

### 5.4.c Eye Contact → Trustworthiness

| Design  | Implementation                                       | Key Concern                                |
|---------|------------------------------------------------------|--------------------------------------------|
| Between | Different participants see different gaze conditions | Individual differences in trust tendencies |
| Within  | Same participants rate BOTH gaze conditions          | Demand to rate them differently            |

**Typical choice:** Within-subjects with counterbalancing

**Reasoning:** If each participant rates multiple faces, we can have them see some faces with direct gaze and some with averted gaze. This controls for individual differences in how trusting people generally are.

## 5.5 Counterbalancing in Within-Subjects Designs

If you use a within-subjects design, you must address order effects through counterbalancing:

**Counterbalancing:** Varying the order of conditions across participants so that order effects cancel out.

- Half of participants: Condition A first, then Condition B
- Other half: Condition B first, then Condition A

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## 6 Part 5: Connecting Design to Statistical Test

### 6.1 The Critical Link

This section is essential for your Research Plan. You must demonstrate understanding of WHY your design choice leads to your test choice.

#### ! Key Principle

The structure of your data (determined by your design) dictates which statistical test is appropriate.

### 6.2 Between-Subjects → Independent Samples t-test

**The logic:**

- You have two groups of participants
- Each participant provides one score
- The scores from Group A are independent of scores from Group B
- We compare the two group means

**Why “independent”?:** The scores in one group tell us nothing about the scores in the other group. Participant 1 in Group A has no connection to Participant 1 in Group B.

**Example:**

“This study uses a between-subjects design with different participants randomly assigned to each font condition. The data therefore consist of two independent groups of recall scores. An independent samples t-test is the appropriate analysis because it compares the means of two unrelated groups.”

## 6.3 Within-Subjects → Paired Samples t-test

### The logic:

- You have one group of participants
- Each participant provides two scores (one per condition)
- The scores are paired — each person's Condition A score is linked to their Condition B score
- We compare the mean difference within each person

**Why “paired”?**: Participant 1's score in Condition A is meaningfully connected to Participant 1's score in Condition B — they come from the same person.

### Example:

“This study uses a within-subjects design with each participant rating faces in both gaze conditions. Because each participant provides paired scores (one trustworthiness rating for direct gaze, one for averted gaze), a paired samples t-test is appropriate. This test analyses whether the mean difference in ratings within individuals is significantly different from zero.”

## 6.4 The Consequences of Using the Wrong Test

What happens if you use the wrong test?

| Error                             | Consequence                                             |
|-----------------------------------|---------------------------------------------------------|
| Independent t-test on paired data | Loses statistical power (harder to detect real effects) |
| Paired t-test on independent data | Violates assumptions, produces invalid results          |

Both errors lead to incorrect conclusions about your research question.

## 6.5 Complete Justification Template

For your Research Plan, structure your justification like this:

“This study employs a [between-subjects / within-subjects] design because [reason for design choice — reference advantages and how disadvantages are managed].

Given this design, the data will consist of [description of data structure — independent groups OR paired scores].

Therefore, [an independent samples / a paired samples] t-test is the appropriate statistical analysis. This test will compare [what is being compared] to determine whether [the research question].”

## 7 Part 6: Method Section Structure

Your Research Plan requires a brief Method section. Here's what belongs in each subsection.

### 7.1 Design

State the design type and define your variables:

"This study employed a between-subjects experimental design. The independent variable was font type with two levels: fluent (Arial, 16pt, black) and disfluent (Comic Sans MS, 12pt, 75% grey). The dependent variable was free recall score (number of words correctly recalled from a 20-word list, range 0-20)."

**Must include:**

- Design type (between-subjects, within-subjects)
- IV with levels fully operationalised
- DV with measurement details

### 7.2 Participants

Describe who participated and how they were assigned:

"Sixty undergraduate psychology students will be recruited through the department participant pool. Participants will be randomly assigned to either the fluent font condition (n = 30) or the disfluent font condition (n = 30)."

**Must include:**

- Target sample size
- Recruitment method
- Assignment method (random, counterbalanced)

**For within-subjects designs**, specify counterbalancing:

"Order of conditions will be counterbalanced, with half of participants seeing direct gaze faces first and half seeing averted gaze faces first."

### 7.3 Materials

Describe everything participants will see or use:

"Stimuli consisted of 20 common concrete nouns selected from the MRC Psycholinguistic Database, matched for word frequency and imageability. In the fluent condition, words were displayed in 16-point Arial (black text on white background). In the disfluent condition, the same words were displayed in 12-point Comic Sans MS (75% grey on

white background). Presentation was controlled using PsychoPy. Participants recorded responses on a blank A4 sheet.”

**Must include:**

- Stimulus details (what, how selected)
- Condition-specific differences
- Measurement tools/materials

## 7.4 Procedure

Describe what happens, step by step:

“Participants will be tested individually in a quiet laboratory room. After providing informed consent, they will be seated at a computer and given standardised instructions to memorise the upcoming word list. Twenty words will be presented sequentially for 3 seconds each. After the final word, participants will count backwards from 50 for 30 seconds (filled delay), then write down as many words as they can remember for 2 minutes. The session will last approximately 10 minutes.”

**Must include:**

- Setting
  - Sequence of events
  - Timing for each phase
  - Key instructions given
  - Total duration
- 

## 8 Part 7: Setting Up Your Results Table

Your Research Plan asks you to create a summary results table **without actual data**. This demonstrates you understand what you’ll measure and how to organise it.

### 8.1 What the Table Should Show

| Condition      | n  | M | SD |
|----------------|----|---|----|
| Fluent Font    | 30 | — | —  |
| Disfluent Font | 30 | — | —  |

Leave M and SD columns empty — you’re showing the structure, not inventing numbers.

### 8.2 For Within-Subjects Designs

The table structure is similar, but you might also show the mean difference:

| Condition    | M | SD |
|--------------|---|----|
| Direct Gaze  | — | —  |
| Averted Gaze | — | —  |
| Difference   | — | —  |

Or present it as paired observations:

| Participant | Direct Gaze | Averted Gaze | Difference |
|-------------|-------------|--------------|------------|
| 1-30        | —           | —            | —          |
| <b>Mean</b> | —           | —            | —          |

## 9 Summary

| Concept                      | Key Point                                                                         |
|------------------------------|-----------------------------------------------------------------------------------|
| <b>Operationalisation</b>    | Turning abstract concepts into concrete, measurable variables                     |
| <b>IV operationalisation</b> | Exactly how you create different conditions                                       |
| <b>DV operationalisation</b> | Exactly how you measure the outcome                                               |
| <b>Two-tailed hypothesis</b> | Predicts difference without direction — use when uncertain                        |
| <b>One-tailed hypothesis</b> | Predicts specific direction — use when theory supports it                         |
| <b>Between-subjects</b>      | Different people in each condition — no order effects but individual differences  |
| <b>Within-subjects</b>       | Same people in all conditions — controls individual differences but order effects |
| <b>Independent t-test</b>    | Compares means from different groups (between-subjects)                           |
| <b>Paired t-test</b>         | Compares means from same people (within-subjects)                                 |
| <b>Design-test link</b>      | Your design structure determines which test is appropriate                        |

## 10 Check Your Understanding

Before the lecture, make sure you can answer these questions:

1. What is operationalisation and why does it matter for research?
  2. For your chosen topic, how would you operationalise the IV and DV?
  3. What's the difference between one-tailed and two-tailed hypotheses?
  4. When would you choose each type of hypothesis?
  5. What are the main advantages and disadvantages of between-subjects designs?
  6. What are the main advantages and disadvantages of within-subjects designs?
  7. Why does a between-subjects design require an independent t-test?
  8. Why does a within-subjects design require a paired t-test?
  9. What happens if you use the wrong statistical test?
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## 11 Further Reading

For additional depth:

- Diemand-Yauman, C., Oppenheimer, D. M., & Vaughan, E. B. (2011). Fortune favors the bold (and the italicized): Effects of disfluency on educational outcomes. *Cognition*, 118, 111-115.
- Edmonds, C. J., & Burford, D. (2009). Should children drink more water? The effects of drinking water on cognition in children. *Appetite*, 52, 776-779.
- Coolican, H. (2017). Chapter 2: Planning research. VLeBooks